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Functional Mapping of Human Motor Areas Using fMRI and K-Means Clustering

NX-421

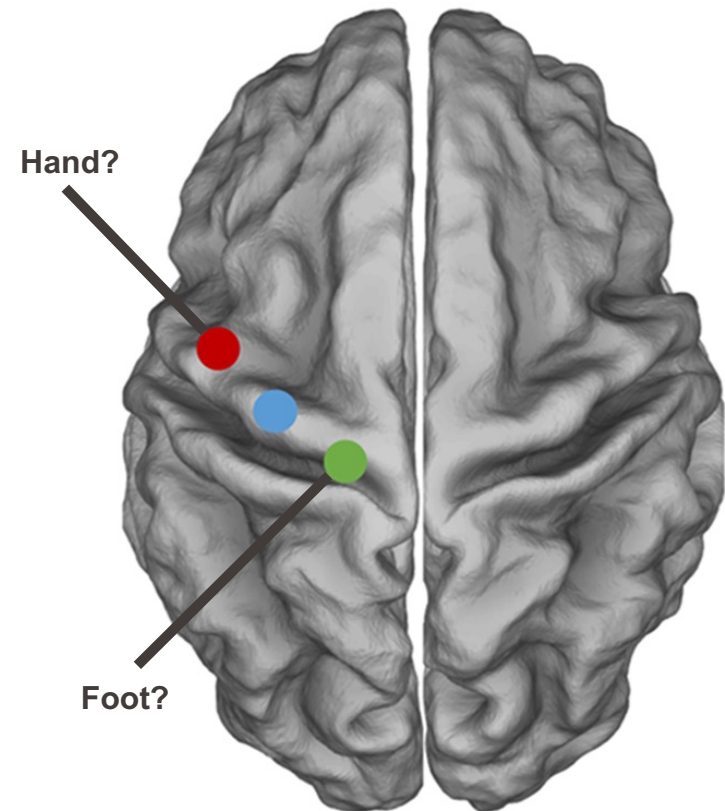
Neural Signals and Signal Processing

Miniproject 1

Fall 2025

Dataset and Motivation

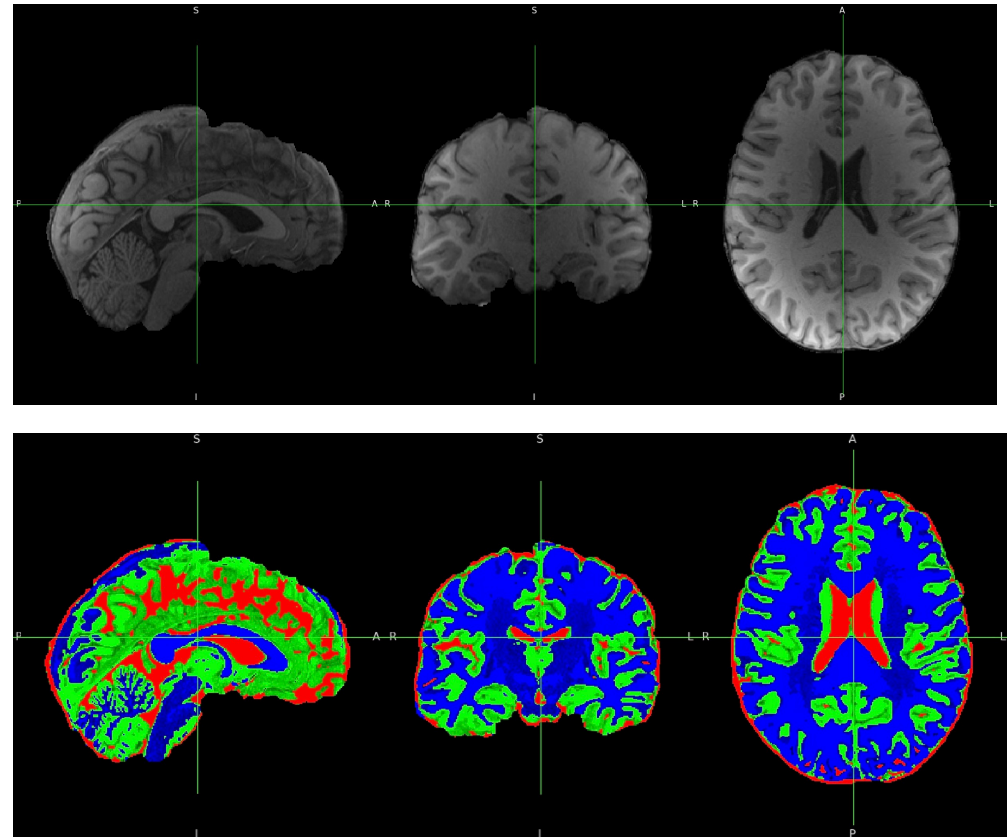
- **Goal:** map motor areas associated with **hands** and **feet** movement
- **Dataset:** subset from *Human Connectome Project* (motor task fMRI)
- **Experimental paradigm:**
 - **Movements:** fingers, toes, tongue
 - **Block design:** 2 runs, 13 blocks each, 3 fixation blocks



Structural preprocessing

Skull-stripping & Segmentation

- **Removal** of skull, scalp, and non-brain tissues from the T1-weighted image
- **Segmentation** into gray matter (GM), white matter (WM), and cerebrospinal fluid (CSF)

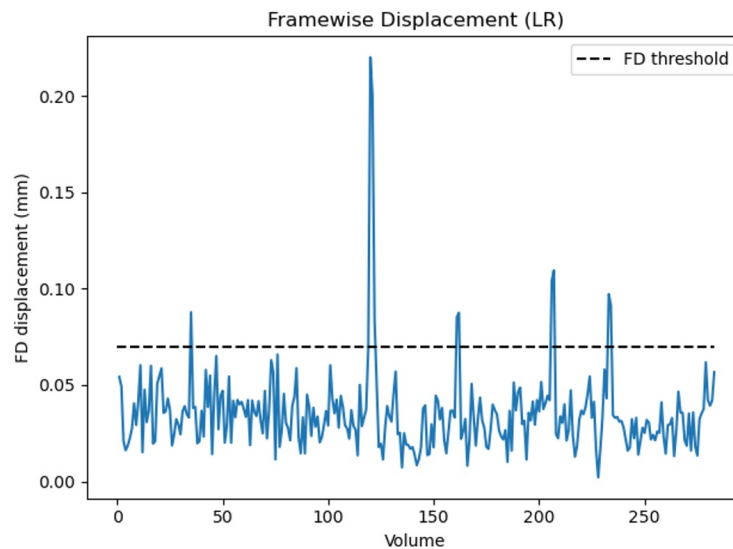


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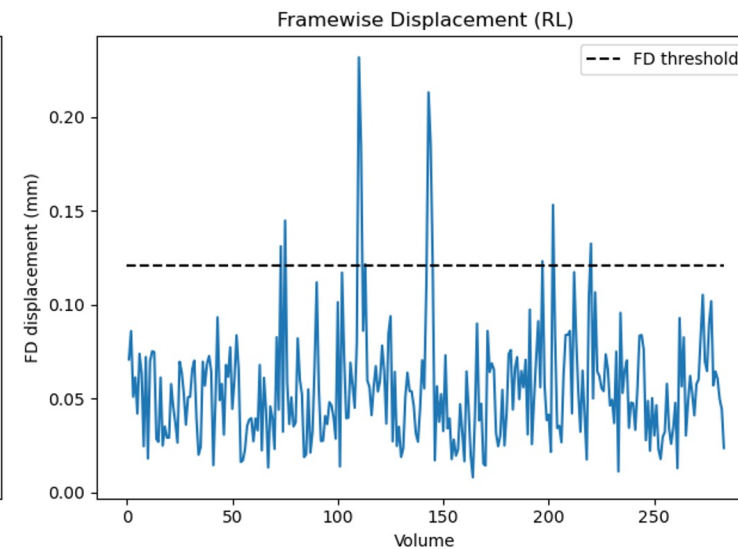
Functional preprocessing

Motion correction & Framewise displacement

- **Head-motion correction** applied separately to each run using a 6-DoF rigid-body transformation
- **Framewise displacement (FD)** computed from estimated motion parameters
Spikes above → frames with excessive motion



Indices of frames above threshold: [35, 120, 121, 122, 161, 162, 206, 207, 233, 234]

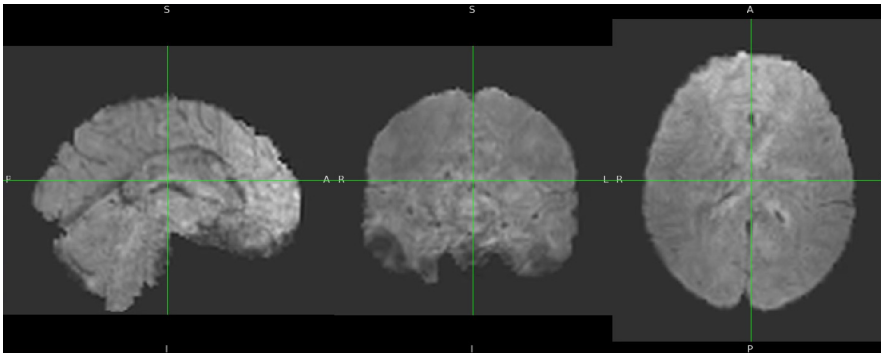


Indices of frames above threshold: [73, 75, 110, 111, 113, 143, 144, 197, 202, 220]

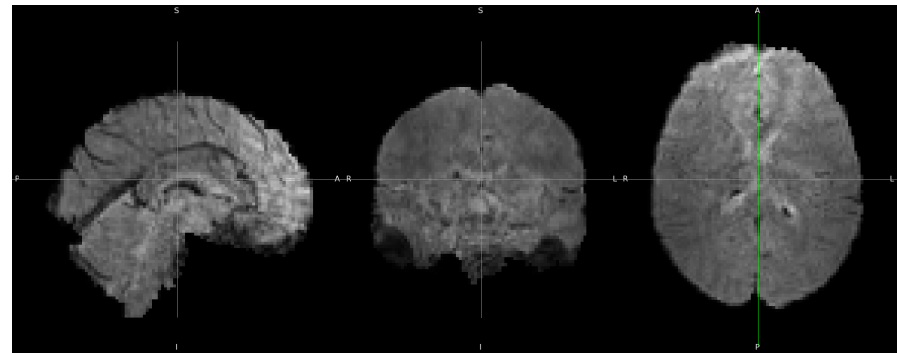
Functional preprocessing

Co-registration & Smoothing

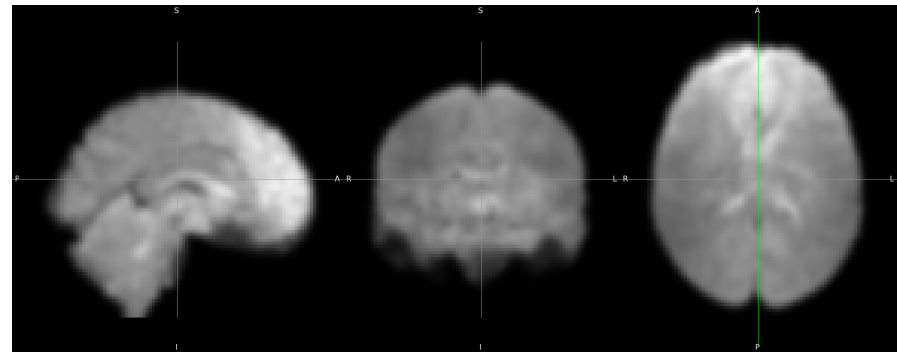
- Functional data is aligned to anatomical data
- Smoothing to improve signal to noise ratio



Before smoothing



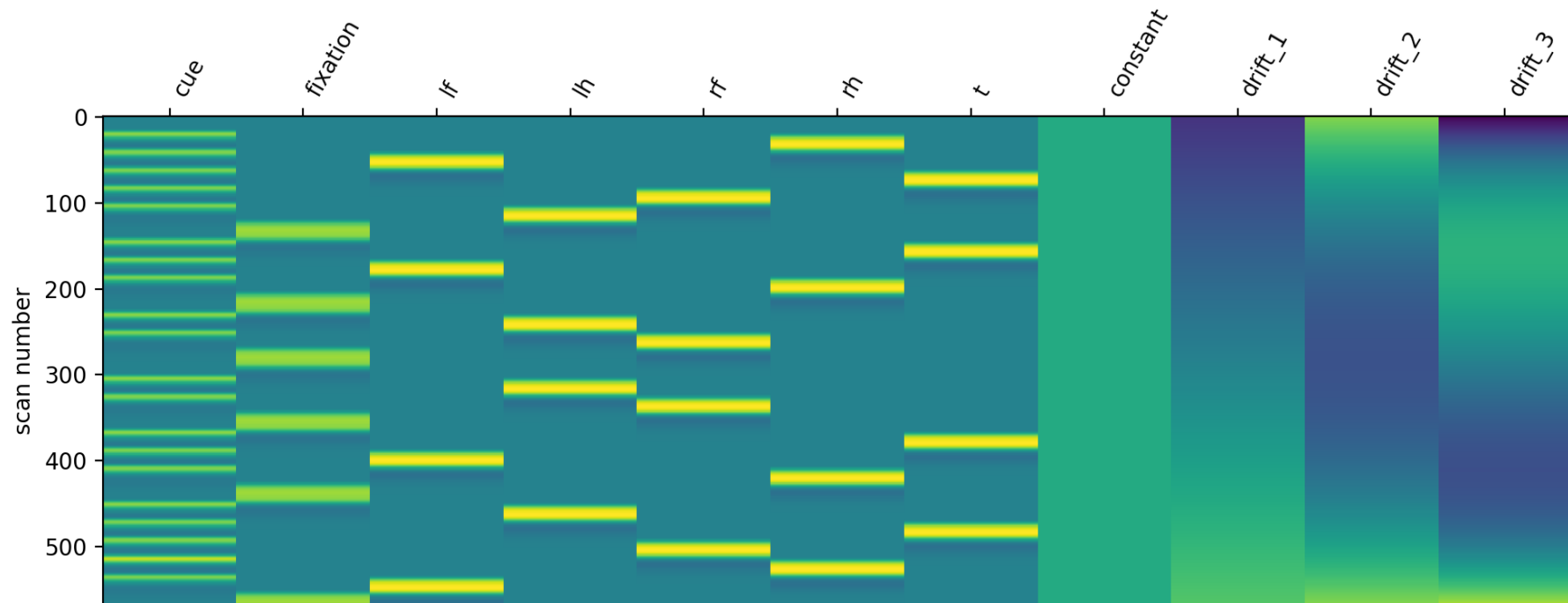
After smoothing



GLM

Design Matrix

- All experimental conditions and third-degree polynomial drift were considered.



GLM

Practical

Contrast: Hand > Foot,

$c = [0, 0, +1, +1, -1, -1, 0, 0, 0, 0, 0]$

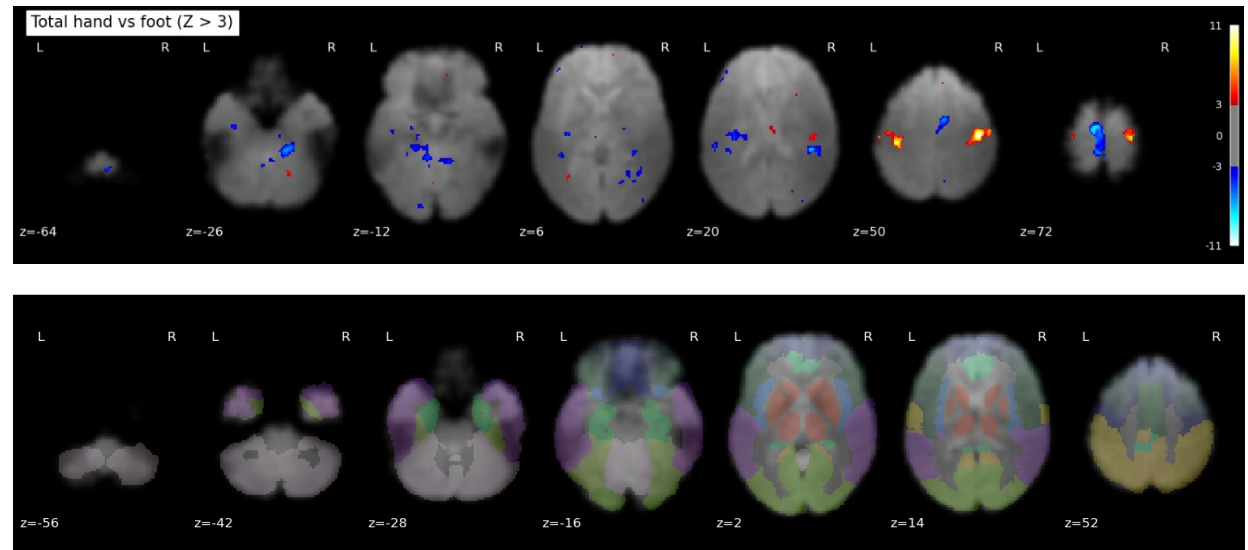
Threshold: $Z > 3$

Maximal activations:

- Contralateral primary motor cortex (M1)
- Supplementary motor area (SMA)
- Ipsilateral cerebellum

Negative activations (Foot > Hand):

- Medial primary motor cortex (leg area)
- Cerebellar vermis



Theoretical-second level analysis

- Second-level (group) analysis estimates **between-subject variability**
- It requires **multiple independent subjects**
- Current dataset: only **1 subject** → **no inter-subject variance**

More individuals and age info

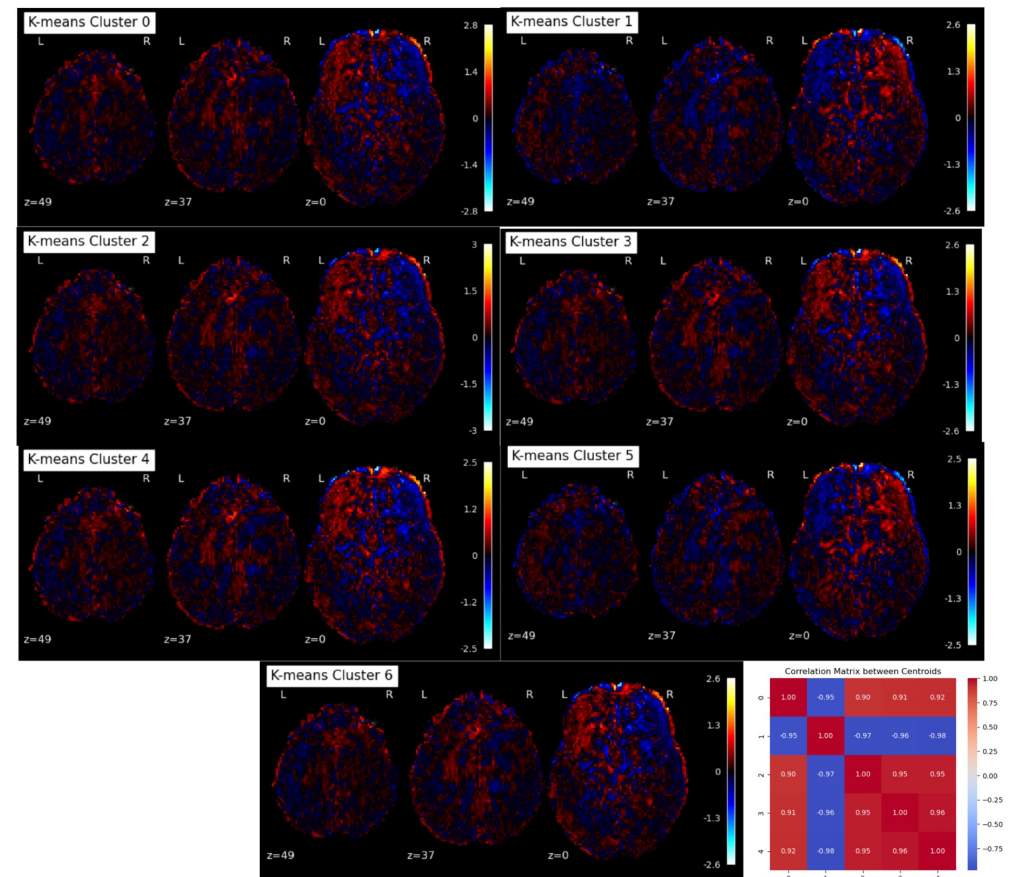
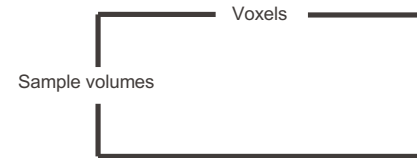
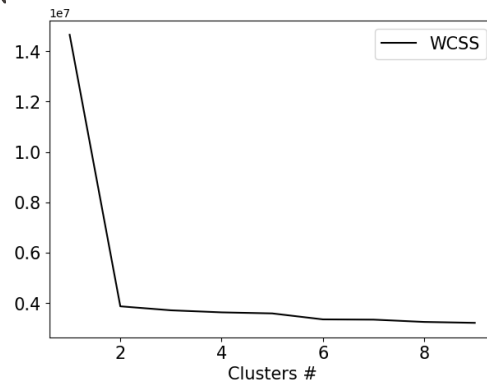
- Each subject's Hand > Foot contrast = **one data point**
- Second-level GLM:
 - Y = contrast estimates (10×1)
 - $X = [1, \text{Age}] \rightarrow$ design matrix
 - β_1 = mean activation across group
 - β_2 = age effect on activation

K-Means Clustering

- **Data-driven clustering** of activation patterns
- **Minimize Within Cluster Squared-Sum** via centroid update

$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

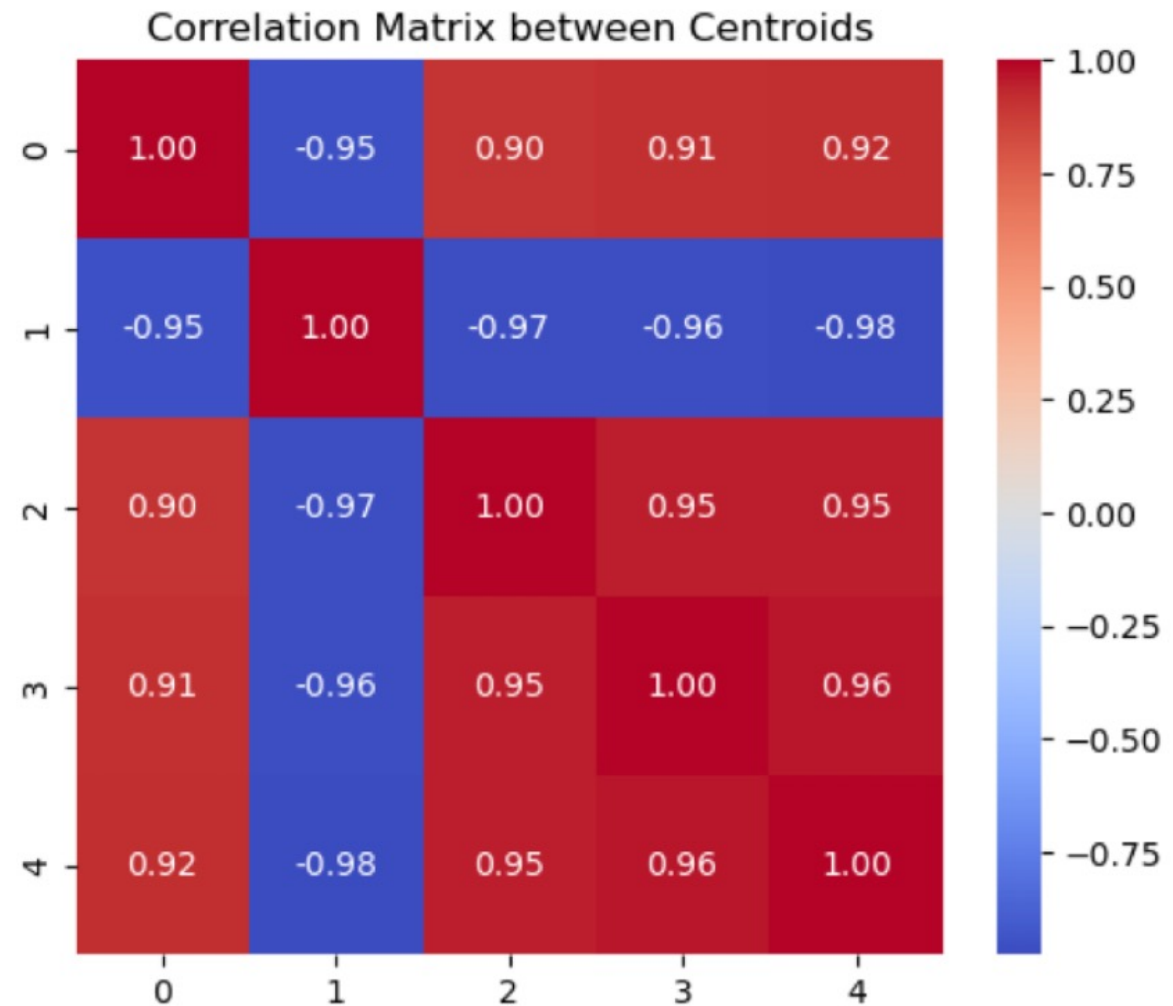
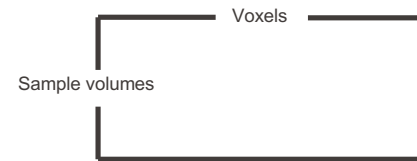
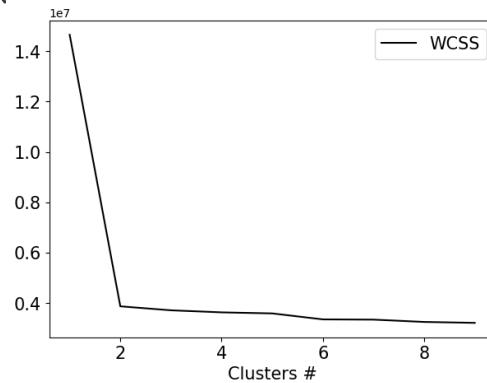
- **7 axial brain maps** matching task conditions (despite elbow = 2)



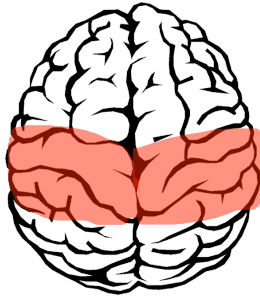
K-Means Clustering

- **Data-driven clustering** of activation patterns
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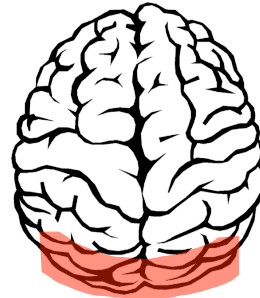
$$WCSS = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$



Functional Networks



Sensory-motor



Visual



Attentional

GLM

- Optimal for **precise localization of activation** in **voxel-wise maps**

MVPA

- **Explores brain states** beyond task design



Author	Sciper	Contributions
Changlan Chen	406114	GLM
Davide Micheletti	386557	Report, K-means
Gianmarco Placì	414573	Skull Stripping, Concatenation, Motion correction, K-means
Vincenzo Riveccio	409465	Coregistration, Smoothing, Design matrix
Pasquale Scognamiglio	416558	Intro, K-means

Thank you for your attention!

NX-421

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